

RELATIONAL TRAUMA IN EARLY CHILDHOOD AND ITS INFLUENCE ON THE
REGULATORY FUNCTION OF THE AUTONOMIC NERVOUS SYSTEM AND
CORRESPONDING PERSONALITY STRUCTURE

A dissertation submitted

by

MICHAEL SUMMERLIN

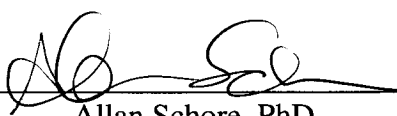
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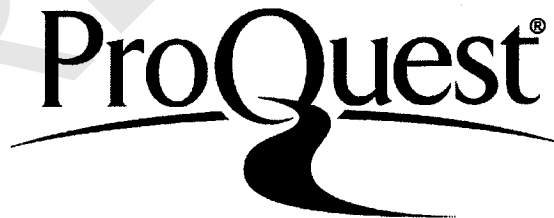
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PREVIEW

ABSTRACT

Relational Trauma in Early Childhood and its Influence on the Regulatory Function of the Autonomic Nervous System and Corresponding Personality Structure

by

Michael W. Summerlin

The purpose of this study is to explore relational trauma in early childhood and its influence on the regulatory function of the autonomic nervous system (ANS) and corresponding personality structure. The method of inquiry will be a theoretical-hermeneutic and correlational design. Relational trauma in early childhood will be generally defined as an impairment of the ANS's ability to regulate one's psychobiological processes while experiencing an environmental stressor. These dysregulated processes consist of cognition, emotive feedback, and affect from one stimulus to the next, as a result of having experienced various degrees of one or more of the following: abandonment, neglect and/or physical abuse during the first two years of human development. Findings of this study suggest that when there has been relational trauma in early childhood, a more rigid and less flexible personality structure is developed allowing the adult in later years to experience the feeling of a more predictable internalized impression of stability (secure base). Furthermore, with the decrease in tolerance for ambiguity comes an increase in dualistic inflexible cognitive function.

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Table of Contents

Chapter 1	1
Introduction	1
Relevance of Topic	1
Chapter 2	9
Literature Review: Projected Dualities	9
Jung's Feeling Function	12
Jung's Feeling Function and Schore's Evaluative System	15
Freud's Dualistic Foundations	16
Jung's Being-Not Being Epistemology	18
Structure and Function	19
Regulation Theory and Its Underpinnings	19
Regulatory Function of the ANS	21
Dual Limbic Circuits	22
Mediating Limbic Components and the ANS	23
Autonomic Influences in Early Development	26
Relational Trauma in Early Childhood	28
Messages From the Past for the Present	31
Projection Identification: Affective Communication	31
Dissociation: The Organisms' Last Resort	32
Borderline Personality Disorder	34
BPD and Neuro-Circuit Construction Theory	36
Chapter 3	40
Methodology and Procedures	40
Interdisciplinary Research Methods	40
Quantitative Research Methodology	41
Participant Organization	41
Participant-Based Quantitative Method	41
Data Gathering	42
Data Analysis	44
Participants	44
Materials	44
Ethical Considerations	44
Chapter 4	47
Results	47
Overview of the Content and Organization of the Findings	47
Actual Findings	49
Discussion of Findings	54
Image-distortion: Idealization – Devaluation; Hypothesis 1	54
Maladaptive: Findings From DSQ Population Means	54
Self-sacrificing: Findings From DSQ Population Means	55
Adaptive: Findings From DSQ Population Means	55
Limitations of Study	56
Statement of Research Question and Findings	57

Chapter 5	58
Clinical Implications	58
A Relational Disorder at Heart	58
Messages in a Body	59
Archetypal Psychology: A Redeemer of Exploration	59
Eros and Psyche: Mythology of a Paradigm Shift	61
Implications of Interdisciplinary Theory	65
Psychotherapeutic Efficacy	65
Paradigm Shift	66
Medical Specificity	66
Regulation Theory and Medical Specificity	67
Recommendations for Future Research	70
Conclusion	70
References	73
Appendix A	81
Appendix B	82
Appendix C	83
Appendix D	84
Appendix E	94
Appendix F	100
Appendix G	102

The style used throughout this dissertation is in accordance with the *Publication Manual of the American Psychological Association* (Fifth Edition, 2001) and Pacifica Graduate Institute's *Dissertation Handbook* (2007-2008).

Chapter 1

Introduction

My interest in this research has been largely influenced by heuristic investigations and Dr. Allan Schore's postulate that the development of an autonomic nervous system (ANS) is experience-dependent and is the central psychobiological mechanism by which we evaluate our respective environments (2003).

When the ANS is unable to regulate, but in effect begins to inhibit an individual's capacity to thrive, then one's ability to evaluate stimuli in the environment is jeopardized (Schore, 2002b). It has been my heuristic experiences with affect dysregulation and the coinciding difficulty evaluating objects in the environment that have lead me to research the regulatory function of the ANS and how relational trauma in early childhood can impair a patient's capacity to effectively regulate and evaluate from one condition to the next. As a clinician, I have been particularly interested in finding ways in which one can recover from the effects of relational trauma. It has been my experience that *psychobiological revision* is most likely to occur in relation to others, just as the injuries that impacted the ANS and its influences on an evaluative system likely occurred in relation to others (Schore, 2003b).

Relevance of the Topic for Clinical Psychology

Each of us is outfitted with an autonomic nervous system, a system that helps to evaluate objects in our environment as beneficial and/or threatening and then acts in response to its evaluation (Bandler, Keay, Floyd, & Price, 2000). It is a psychobiological system; therefore our thoughts and ideas are fused with our

experience of the external in a way that doesn't permit separation between the internal and external (Platt, 2007). Ways in which this phenomenological arrangement challenges clinicians and patients attempting to create *psychic space* (Bion, 1965) are explored in chapter 5.

In coming chapters, I put forth examples by which problematic personality organization results from environmental stimulation that is experienced as threatening, thereby bringing about what I have observed to be concretized psychobiological mental constructs that jeopardize one's relational life. Additionally, James Hillman's (1983) concept for *deliteralizing* the ideas that make up a patient's seemingly fixed mental constructs are explored as an overarching philosophical approach to treatment with borderline personality disordered patients (BPD). A broad-stroke description of Hillman's approach could be described as analogous to looking through lenses, taking them off and investigating what is being seen through the lenses, and then reflecting on how they color the intrapersonal experiences of various stimuli. In conjunction with Hillman's theory is my position that there are inherent advantages for clinicians that consider the common denominator of biology, one of which is a conceptually unifying effect for psychotherapists that are working with culturally and ethnically diverse populations. Utilizing the research that has recently come out of neuroscience affords clinicians a broad-spectrum perspective that other theoretical approaches to psychotherapy do not hold. As for the relevance to clinical psychology, it seems self-evident that Schore's interdisciplinary concepts are applicable to this researcher/clinician and all others that have an understanding of the basic working components of an ANS.

Regarding the most severe cases of psychobiological relational injury, it could be said that we come to represent what hurts, and what heals, and with this in mind, there lies the boon that serves to motivate this researcher to explore various treatment interventions. As a result, this dissertation examines two-person models of psychotherapy formulated to address relational trauma and corresponding personality disorders. Additionally, metapsychology is utilized to offer an overarching perspective so that the influences of larger systems can be considered as treatment approaches are evaluated. For example, as this researcher peers through the lenses of metapsychology, it seems as though clinicians are drawn to serve because on some level of consciousness one feels built up by the act of helping. At a microbiological level, the clinician is intrinsically rewarded along the way by reinforcing dopaminergic signals received from ANS structures acting to reinforce continued exploration of our human condition. And at a societal level, others benefit as a result of our behavior (Bono & Judge, 2004).

Evaluation that coincides with an individual's explorations is derived from a storehouse of organically held data that has accumulated through prior experience combined with stimuli from the present environment (Hebb, 1949). The messages often act to propel one into action, a state of arrest, or in some situations the individual is able to assess further, process longer, and then respond (Keay & Bandler, 2001). My bias is that an individual will respond in a way that is more conducive to thriving if one is more attuned with his/her own internal psychobiological processes. That is, if one is in a psychobiological state that comprises a more integrated processing of one's self state (i.e., internal stimuli, external stimuli, and the

coalescing of the two). In forthcoming sections of this document pertaining to physiological structure and function, I provide findings that show the informative salience of those processes. Because we are psychobiologically affected in the presence of another and vice-versa, in dyad we enter into a state of emotional regulation, dysregulation, or a variant state associated with regulation/dysregulation, which is largely fostered by each others' affective communication (Skolnick, 2006). A short list of mediums for affective communication consists of facial expression, duration and intensity of eye contact, volume of voice, prosody of voice, and body language (Schore, 2001a).

In the following paragraphs, I explain how in viewing our selves as systems of evaluation for survival, the lines that often divide us dissolve and we are subsumed into the category of human being. As a result of this, we gain a profound understanding of human nature thereby opening up immediate potential for effective clinical applications. As mentioned earlier, this research topic is grounded in data pertaining to psychobiological structures and processes that are common to all. As a result of being familiar with these mechanisms and/or processes, we better understand the origins of psychopathology, substance abuse, domestic violence, and various other areas of clinical focus (Schore, 2001a).

It is widely accepted in the field of psychology and other disciplines that a basic tenet of being human is having an innate drive to perpetuate life—our own and/or the lives of others that we feel attached (Lorenz, 1981). As we go about living we find ourselves drawn to certain things in our environment that we have evaluated as beneficial and feel can build us up, subsequently perpetuating life (an account of the

developmental underpinnings of this is detailed in an upcoming chapter). For example, most people are drawn to possessing adequate shelter that offers a structure that will keep them safe from things in the environment that they perceive could potentially tear them down. Efforts made by people in the name of that goal are many. Some work tirelessly in jobs that are not satisfying in order to afford to buy that feeling of security. Conversely, there are things in our environment that we find threatening and as a result we avoid what we have evaluated as something that could tear us down, such as a cold rain that the shelter from a house might protect us from (Maslow, 1970). It has been my experience that the process of evaluation is sometimes conscious and sometimes out of the realm of immediate awareness; this process is explored in coming chapters.

Research coming out of the field of affective neuroscience provides empirically based evidence that can enable clinicians to view the patient's presenting symptoms as indicative of the point of human development at which the relational trauma initially occurred. Furthermore, provided with data that holds a description of pathogenesis, the clinician can hold a more grounded understanding of their experience of countertransference and what it might mean within the context of the therapeutic relationship. This grounded understanding comes by way of research that has uncovered the various patterns of preverbal communication along with the ontological challenges that begin in infancy and persist throughout the organism's development (Kohut, 1984).

Although an understanding of the importance of experiencing affective communication is well established, as clinicians and theorists we are often drawn

toward *autoregulating* effects that can come from categorizing and assessing while interacting with a patient in treatment. As a result, salient communication can be missed due to what I suggest is the therapist's attempts to quell the effects of their own psychobiological dysregulation through hyper-categorization (a left hemispheric function of the cortex).

Another explanation of how right hemispheric affective communication can be missed comes by way of contemporary theorists Bateman and Fonagy (2006), who advise clinicians to avoid causing "too many iatrogenic effects as the *attachment system* is activated" (p. 36). It is my view that clinical focus on what is "too many" or not enough often results in not experiencing the transference/countertransference intersubjective field (Stolorow, 1995). I suggest that focusing on what might be "too many" coincides with counting, which can disconnect the clinician from a source of affective communication. In coming chapters I discuss how placing too much emphasis on sequencing is primarily a left hemispheric function and can mitigate the ability to understand and process affective cues. Furthermore, an attachment rupture, in some cases, is often a necessary precursor to a corrective emotional experience (Winnicott, 1971).

Affective communication is known to emanate from the right hemisphere of the cortex, which comprises the human being's overall condition most comprehensively (Schoore, 2003). In general, the right hemisphere serves the purpose of integrating one's external sensory impressions (exteroceptive) with internal (interoceptive) impressions, collectively contributing to one's state of consciousness, a source for affective communication conveyed through an intersubjective transference field

(Schutz, 2005). When a clinician is overly focused on assessment and interpretation, vital affective communication can be missed, as assessing and interpreting frequently put the clinician at an emotive distance that renders the therapist out of reach from a critical affective-communication field during interaction.

The idea that one must be on constant guard not to “damage the patient” suggests an approach to the relation that is fragmented and arrhythmic in its composition. Hutterer and Liss (2006) suggest that the clinician would be well advised to consider their own feelings of hesitancy regarding an affective exchange with the patient, as “traumas that have occurred early in life . . . may be inaccessible to words” and are consequently communicated and understood through felt sensations (p. 297). Clinicians have often gone to great lengths in order not to expose themselves to becoming emotionally overwhelmed with countertransference (Maroda, 1999). There are a number of theoretical approaches that prescribe objectivity or simply leave out relational dynamics that can act to inform the clinician and patient. Maroda suggests whereas relational approaches to the practice of psychotherapy have become more prevalent in the field over the past decades, there still remains an inescapable guardedness that is perpetuated by all who practice. I suggest that this perpetuation is primarily due to both psychological and organic constructs.

In my view, a constant challenge for clinicians is to maintain an awareness of imbedded mental constructs that border on James Hillman’s definition of psychosis. These constructs seem to soften the sharp edges of a world that would otherwise trigger our *strong attractor structures* (SAS); (Amini, Lewis, Lannon & Louie, 1996) and leave us in a state of psychobiological discomfort. Hillman (1983) suggests that it

is an aspect of the culture of the field of psychology that its shadow be concealed from those of us that are a part of the field or are being served by it. It is my position that the fundamental fear of being torn down is so profound that the very institutions that profess to resolve the most complex of human dilemmas—are not immune. Institutions and their corresponding ideals are just as vulnerable as the people that created them. Barriers to seeing the field of psychology's shadow are imbedded within the fabric of the concepts and potentially impair professionals as they attempt to assist patients. This document explores some of the psychobiological underpinnings of that shadow in an effort to afford clinicians and their patients the ability to navigate through unconscious problematic relational patterns.

PREVIEW

Chapter 2

Literature Review: Projected Dualities

Western civilization's first treatise on physiology and psychology by René Descartes still influences how practitioners and researchers approach their work (Wozniak, 1995). However, Hughlings Jackson's term *physiological bottom of the mind* (autonomic nervous system) did much to bring the body into theories of mind (1931). The ideological separation between the concept of mind and the concept of body that ensued in western civilization subsequent to Descartes' treatise on the mind/body question currently persists through the theories and practice of western science and medicine (Dobson, 2001). As it relates to the treatment of relational trauma incurred during early childhood, the conceptual separation of mind and body contradicts recent evidence that shows a physiological foundation out of which thoughts manifest. In Descartes' famous doctrine *Cogito ergo sum* (I think therefore I am), he seems to suggest that the activity of the mind supersedes physical existence. The residual effect of Descartes' doctrine can be found in some cognitive-based approaches to psychotherapeutic interventions, such as those that address depressive symptom control through *changing thoughts* (Beck, Rush, Shaw, Emory, 1979).

While approaches like "changing thoughts" have been championed by psychologists, clinicians, and patients, there have concurrently been various psychodynamic-based theories to suggest that psychobiologically based drives predicate one's thoughts and behaviors. And finally, contemporary neuropsychanalytic theory suggests that there is no time-bound sequence that applies to thoughts, feelings, and behaviors, but rather a non-linear occurrence that equals a

state of consciousness (Marks-Tarlow, 2008). Some contemporary theorists from various schools of thought have moved to an idea of consciousness that suggests that there is no real separation between unconscious or conscious mind, but rather a state of being that cannot be well defined by either term. Wilkinson (2004) states the following:

Gone forever are the unquestioning days of the dualism of Descartes, when the mind and brain could be understood as two entirely separate entities and scholars of one would not have been expected to be acquainted with the scholarship of those who studied the other. (p. 83)

I suggest that dualism, as noted by Wilkinson, continues to exist as evidenced by conceptualizations such as conscious and unconscious. In short, it seems theorists continue to divide and draw lines, yet also dissolve the conceptual lines in an effort to integrate, only to have the lines drawn again to relieve the discomfort that comes from what Jung (1953) called “the tension of the opposites” (p. 23). It seems the degree of permanence that is assigned to the lines that are drawn becomes proportional to the pathological dynamics of the psyche in question. This matter will be explored through experiment in chapters 3 and 4.

During the first half of the 20th century, a historically salient shift away from Descartes’ idea of cognitive primacy began with neuroscientist Hughlings Jackson’s (1931) term for the autonomic nervous system, “the physiological bottom of the mind”. In the view of this researcher, Jackson’s assertion regarding the ANS began an ongoing exploration in the human sciences of a new frontier that has eventually brought about an ideology that depicts a state of being that is neither conscious nor unconscious, and neither mind nor body, but an inseparable fusion of organic structure and mental function. Philosopher Martin Heidegger seemed to refer to this

state of being as synonymous with experiencing an *event horizon*, a hermeneutically sponsored term that takes one's presence and history into account resulting in an interpretation of the experience of being (Palmer, 1969).

A central goal for this work is to find the ties that bind the regulatory function of the ANS with empirically supported theories on psychological structure, in order to formulate more effective psychotherapeutic treatment interventions. Through this process of research, I have come upon many theorists who have assigned various names to what I view as an internal dichotomous energy management system consisting of two paralimbic structures. For the purpose of supporting my postulate regarding the existence of dual organic structures that manifest into survival-based dualistic psychological defense systems, I have selected Freud and Jung's theories in order to expand on how their psychologically based terms for the dual circuits represent central organizing principles within each of the theories. In this literature review, I discuss my view that Freud and Jung's respective theories were dualistically-based approaches that have stood the test of time as a result of their organic and ubiquitous nature. Additionally, I will utilize Allan Schore's regulation theory to provide connecting points between physiological structure and psychological organization and function.

I begin by covering Carl Jung's (1953) assertion that each individual is faced with a constant survival question of being or not being, and discuss how Jung's *feeling function* addresses the human being's survival-based needs. Secondly, I offer Sigmund Freud's use of Eros and Thanatos as another example of how human beings are in a perpetual state of navigating for survival. These two forefathers of depth

psychology took very different courses of exploration throughout their lives, although their respective theories both held that ultimately the human organism is always faced with living and dying, and it is out of this grounding dualistic starting point that their musings sprung forth.

By examining anatomical structures and corresponding developmental processes, a *proof-system* (Schore, 2003a) can be developed that supports and expands on these established theories that depict psychological structures and coinciding psychodynamics. In recent neuro-scientific research, dual paralimbic circuits are shown to be central to affect regulation (Swain, Lorberbaum, Kose, & Strathearn, 2007). I suggest that they are also foundational to the development and perpetuation of personality structure.

It has primarily been Freud and Jung's theories that have enabled me to get to this point of discovery, and with those concepts I would like to use a psychoneurobiologically based theoretical approach to act as a substrate to further validate what largely came about through Freud and Jung's abstracts.

Jung's feeling function. Our experience of a being/not-being split is biologically predetermined by an evaluative survival system (i.e., autonomic nervous system) that paradoxically motivates our cognitive efforts to rise above it. Jung (1953) describes such conscious cognitive efforts: "The essence of the conscious mind is discrimination; it must, if it is to be aware of things, separate the opposites" (p. 25).

Jung described the feeling function of the personality as having an evaluative role. The feeling function is one of four aspects of personality that Jung included in his typology of self (Jacobi, 1944). In *The Psychology of C. G. Jung*, Jacobi uses a

graphic illustration (Appendix A) of the Chinese *Taigitu* symbol, (yin/yang) which shows the feeling function placed within the dark shaded half of the symbol (p. 13). Graphically, this location for the feeling function within the symbol spatially and topographically parallels right hemispheric lower cortices. It seems that Jung instinctively understood that a significant aspect of consciousness was outside of conscious awareness. He also understood that previous generations had projected this understanding which came to take on symbolic form. Jung's claim to empiricism held symbols and beliefs as facts in and of themselves. In *Psychology and Religion* (1938) he asserts the following:

Speaking for instance of the motive of the virgin birth, psychology is only concerned with the fact that there is such an idea, but it is not concerned with the question whether such an idea is true or false in any other sense. It is psychologically true in as much as it exists. Psychological existence is subjective in so far as an idea occurs in only one individual. But it is objective in so far as it is established by a society—by a consensus gentium (p. 3).

Given Jung's quote, whereas symbols are not indicative of any one truth, it seems a symbol can be linked back to matters of fact that brought on the formulation of the symbolic form in question. I suggest here that the *Taigitu* symbol is a representation of co-existing opposites within our biology, with the light side of the symbol representing biological structures that manifest the release of energy and the dark side of the symbol depicting the organism's corresponding unknown reserves. Additionally, the graphic of the *Taigitu* contains a spot of dark in the light half and a spot of light in the dark half; thereby each side of the mandala symbol contains a portal to the opposite side giving the symbol an ever-flowing, non-linear dynamic quality. The corresponding psychobiological functions that are held by the *Taigitu*

entail an individual going from a parasympathetic state of hypoarousal (avoidance) to a sympathetic state of hyperarousal (approach) in an ongoing alternating pattern, which are an organism's two basic modes of energy regulation (Recordati, 2003). I suggest that this symbol's design stems from an individual and collective unconscious projection of a psychobiological synergistic state.

Even an atomistic view that peers well beyond observable organic matter reveals an uncanny similarity to one's energy management system as well as interpersonal energy dynamics as they apply to spatial negotiations between individuals. In example, physicist Richard Feynman (1963) states the following:

If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis that all things are made of atoms—little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another. In that one sentence, you will see, there is an *enormous* amount of information about the world, if just a little imagination and thinking are applied. (p. 4)

It has been my experience that, considering individual psychotherapy, Feynman's quote provides a profoundly illuminating depiction of the process of "*space negotiation*" (A. Panajian, personal communication, March, 7, 2009) that occurs among relating individuals. In short it seems that, like atoms, human beings attempt to avoid permanent isolation while attempting to maintain associations at what is felt by each individual to be a satisfactory distance. It has been my observation that even patients who appear to have the desire to completely merge with another at some point attempt to make space for themselves periodically to decrease the duration of an intolerable psychobiological state of not-being (Jung, 1953).

Jung's feeling function and Schore's evaluative system. Jung describes the feeling function as having an evaluative quality. For Jung, his definition was not so much about feelings in the traditional meaning, but more about sensing what is beneficial in the environment as well as what could be harmful to the organism (Jacobi, 1944). More recently, Schore (2003) has targeted the ANS as an evaluative system that operates beneath our conscious awareness. The intersections between Schore's evaluative system and Jung's feeling function are primarily threefold: (a) both are described as functioning beneath conscious awareness; (b) both perform vital evaluations of the organism's environment; and (c) each are foundational to the development of personality structure (Jacobi, 1944; Schore, 2003).

With what Schore terms experience-dependent maturation of the evaluative system, Schore's regulation theory offers an extensively researched concept for what I see as the developmental genesis for Jung's feeling function, perhaps Schore's theory identifies the sought-after "Jungian baby." One of the criticisms of Jungian theory has been that it lacked developmental explanations of how early stages in life impact adulthood. Schore's concept of an evaluative system provides a developmentally based etiology of Jung's feeling function. Through interdisciplinary research, Schore has been able to provide a theoretical bridge that offers concepts that cover areas of exploration that were previously uncharted by virtue of the inherent restrictedness of the methodologies and the theorist's idiosyncratic perspective. In the prologue to Jung's biography, Jung (1963) describes how he has experienced life. In my view, his description provides clues as to how his theory of individuation could be expanded by a researcher with the benefit of having Jung's work to consider against the backdrop

of an interdisciplinary modality.

My life has been singularly poor in outward happenings. I cannot tell much about them, for it would strike me as hollow and insubstantial. I can understand myself only in the light of inner happenings. It is these that make up the singularity of my life, and with these my autobiography deals. (p. 1)

As could be understood from Jung's quote, he and his contemporaries were largely theorizing out of their own experiences of living. It would be years before John Bowlby (1969) would actually observe children in the earliest stages of human development, and when he presented his findings, they were met with considerable resistance from the analytic community. From the standpoint of this researcher, Bowlby's work appears to represent some of the base materials that Schore has used in bridging theoretical disciplines and answering questions that clinicians and researchers have long asked. Interdisciplinary theoretical research has provided new guideposts that have served to direct contemporary researchers toward a middle territory that seems to represent a crossroads between imagery and organicity. As a result, it appears that the images and corresponding ideas that led theorists like Jung and Freud to their respective hypotheses were grounded in the body and are of the body, and most recently are resulting in a paradigmatic shift regarding clinical concerns.

Freud's dualistic foundations. Freud, (1922/1950) used the mythological names Eros and death instinct (Thanatos) as terms to represent the individual's core poles which generated psychic influences on the self. In *Beyond the Pleasure Principle*, Freud discussed how his *pleasure principle* concept (Eros) is balanced out by his *reality principle*, (death instinct, Thanatos) by having an inhibiting affect on the former.

Under the influence of the ego's instincts of self-preservation, the pleasure principle is replaced by the reality principle. This latter principle does not abandon the intention of ultimately obtaining pleasure, but it nevertheless demands and carries into effect the postponement of satisfaction, the abandonment of a number of possibilities of gaining satisfaction and the temporary toleration of unpleasure as a step on the long indirect road to pleasure. (p. 7)

I suggest that Freud projected his own autonomic experiences into a concept that he explained with the use of the two terms Eros and Thanatos—with Eros representing sympathetic arousal and Thanatos representing parasympathetic down regulation or withdrawal. In the following, Freud (1922/1950) continues to use the terms pleasure principle (Eros) and death instinct (Thanatos) as he describes the functions of the two branches of the ANS.

The pleasure principle seems actually to serve the death instincts. It is true that it keeps watch upon stimuli from without, which are regarded as dangers by both kinds of instincts; but it is more especially on guard against increases of stimulation from within, which would make the task of living more difficult. This in turn raises a host of other questions to which we can at present find no answer. We must be patient and await fresh methods and occasions of research. (p. 77)

Through data provided by technological resources that were not available to Freud, contemporary researchers (Schore, 2003a) have found the base regulatory function of the ANS (autonomic nervous system) to be arousal inhibition or as Freud stated in the preceding quote, “a guard against stimulation from within.”

While Freud and Jung laid the groundwork for concepts that other theorists studied and expanded on, they were not privy to the technology that has brought the paralimbic circuits to light. As a result of neuro-imaging technology, not only do we know that the circuits exist, we know more about their coinciding functions with affective communication and autonomic responses to environmental stimuli (Swain,